Bifurcations and Chaos in the Lorenz 96 Model

Aditi Ajith Pujar

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1 Introduction

The Lorenz-96 model is a widely used test model for a broad range of fields - from data assimilation and predictability to studies in spatiotemporal chaos. It is often used to test new ideas in various fields, especially in certain machine learning algorithms for parameter learning[1].

Much of this wide range of applications is due to the simplicity and ease with which L96 models can be simulated. Indeed, the Lorenz-96 models were first proposed by the celebrated meteorologist and mathematician, Edward Lorenz (in 1995), *not* to be a realistic model of the atmosphere, but simply "formulated (it) as one of the simplest possible systems that treats all variables alike and shares certain properties with many atmospheric models".[2]

There are different classes of L96 models; the *monoscale Lorenz 96 models* have a single time scale and they are the focus of this project.

1.1 The Monoscale Lorenz 96 Model

The equations of the L96 model are cyclically equivariant. Thus, for a given dimension n, all the equations can be written as:

$$\dot{x_j} = x_{j-1}(x_{j+1} - x_{j-2}) - x_j + F, \qquad j = 1, 2, \ldots n$$

with "periodic boundary conditions", i.e.

$$x_{j+n} = x_{j-n} = x_j$$

The quadrative terms are *advective* and conserve the energy analogue ($\sim \frac{1}{2}\dot{x}^2$), the linear terms are *dissipative* and F is an external forcing parameter (usually $F > 0$ and pumps energy into the system). The smallest n for which the equations are non-trivial is $n = 4$. Thus, all the analysis henceforth is for models of dimension $n \geq 4$.

2 Literature Review

Due to its relative simplicity, we can make some conclusions about topology of the model by analytics. This guides our numerical experiments thus:

2.1 Overarching Features

1. For any dimension n, any $(n-1)$ sphere with radius $\geq r_{critical}$ serves as a "trapping region" for a trajectory. $\implies$ none of the trajectories run away to $\pm\infty$, but remain trapped within some phase volume.

2. $x_F = (F, F, \ldots, F)\epsilon\mathbb{R}^n$ is a trivial equilibrium point for a given value of F. It is the organising centre for much of the system's dynamics. Exploiting symmetries of system, eigen values of the Jacobian of the ODEs can be analytically found at x_F, providing insight about its stabilty. ($Re(\lambda) > 0 \implies$ instability and vice versa)

2.2 Bifurcations

A bifurcation occurs when a small, smooth change in parameters of a dynamical system lead to sudden changes in the topology of the system's flow (i.e. in its overall qualititative behaviour).

Bifurcations could lead to a change in the stability or the number of fixed (equilibrium) points or periodic orbits. The nature of the bifurcations can often be deduced by change in the eigen values $(\lambda)s$ of the fixed points[1] undergoing change. We encounter 2 main types of bifurcations here:

1. **Hopf bifurcations**: for a conjugate pair of eigen values, the sign of $Re(\lambda) = 0$ changes while $Im(\lambda_j) \neq 0$. Thus, the fixed point changes stability; moreover a limit cycle (an 'isolated periodic orbit' is either created or destroyed.)
   If $Re(\lambda_j)$ changes sign from -ve to +ve, a limit cycle is created (called supercritical Hopf bifurcation). If two pairs of eigen values undergo bifurcation for same value of F, it is called a Hopf Hopf bifurcation.

2. **Supercritical pitchfork bifurcation**: a stable fixed point changes turns unstable beyond F_{bif}, giving rise to two new stable fixed points in the process.

In the L96 system: From [3] it is concluded that:

1. For $F > 0$: x_F undergoes either a Hopf or Hopf-Hopf bifurcation (could be supercritical or subcritical). These can be detected by analysing the eigen values of $J(x_F)$.

2. For $F < 0$:

   (a) if n is odd: x_F undergoes a Hopf bifurcation

   (b) if $n = 4k + 2$: x_F undergoes a supercritical pitchfork bifurcation (denoted by 1-PF). The resulting 2 stable branches undergo a supercritical Hopf bifurcation at the same value of F.

   (c) if $n = 4k$: x_F necessarily undergoes 2 supercritical pitchfork bifurcations and then the resulting 4 stable branches undergo a supercritical Hopf bifurcation at the same value of F.

[1] By this, we mean that we linearise the dynamical system $\dot{x} = f(x)$ to $\dot{x} = Ax$ about the fixed points $x*$ (i.e. $A = Df(x*)$), nand the eigen values of A are analysed

2.3 Chaos

Chaos is defined as a *'sensitive dependance on initial conditions'*, i.e. in chaotic systems, two trajectories with arbitrarily close starting point may significantly diverge from each other in finite time.
This is captured by quantities called *Lyapunov exponents*:

2.3.1 Lyapunov Exponents

if $\delta|Z(0)|$ is the initial separation between two trajectories in phase space, and $\delta|Z(t)|$ is the separation at some time $t > 0$, it the two can be related as:

$$\delta|Z(t)| \sim e^{\lambda t}\delta|Z(0)|$$

1. If the system is "linear", i.e.
$$\dot{x} = Ax + f(t)$$
the Lyapunov exponents are just $Re(\lambda)$s, i.e. the eigen values of A.

2. In case of non linear systems, the notion of Lyapunov exponents is a bit more involved. They are defined from the Jacobian matrix
$$\begin{aligned} \dot{x} &= f(x) \\ \dot{Z} &= J(x)Z \qquad \text{(Linearisation)} \end{aligned}$$
Here $J^t(x_0) = \frac{df^t(x)}{dx}|_{x_0}$ is the Jacobian at a point x_0 at time t. It helps capture how a small change at the point x_0 propagates to the final point $f^t(x_0)$.

   Here, Lyapunov exponents are ultimately defined as the eigen values of the matrix Λ where
$$\Lambda = \lim_{t\to\infty} \frac{1}{2t} \log(J^t.Transpose(J^t))$$

For non linear cases, if λ is (the largest Lyapunov exponent)and if $\lambda > 0$, the two trajectories will diverge, as expected for a chaotic system. Thus, for a working definition of chaotic systems, we can say

$$\boxed{\text{If } \lambda > 0\text{, system considered chaotic}}$$

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Chaotic behaviour also appears seemingly random, despite a system being deterministic. Generally, chaos arises in systems with many unstable manifolds - these cause a trajectory to rapidly 'move away' from them, leading to erratic motion similar to pinballs rattling about a machine. There are various different ways multiple unstable manifolds appear, thus, there are different 'routes to chaos' for any system.

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Therefore, the remaining parts of the report are divided into 2 parts: numerically verifying the bifurcation structure outlined above, trying to find the point beyond which behaviour becomes chaotic as a function of n as well as identifying some routes to chaos.

3 Bifurcation Structures

The fixed points had to be found as a function of F to obtain the above bifurcation structures. This was done thus:
ALGORITHM: Root Finding in n-dimensional space

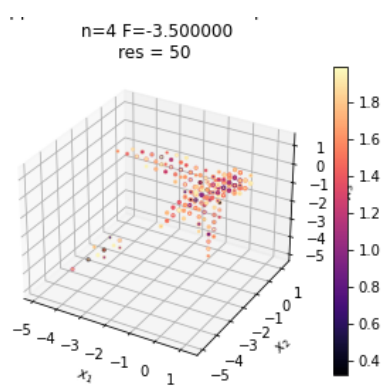
1. Coarse-grain the space, taking some sample points in each dimension. Calculate $||f(x)|| = ||(\dot{x_1}, \dot{x_2}, \ldots, \dot{x_n})||$ at each of the points in the n dimensional grid. Locate and plot only those with $||f(x)|| <$ some small value, ϵ.

2. Taking those x with $||f(x)|| < \epsilon$ as initial guesses, roots are found using n-dimensional Newton Raphson method. Their eigen values are found to understand the nature of the roots.

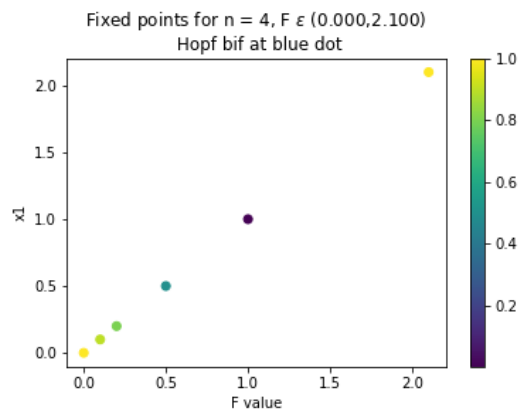
3.1 $F > 0$

3.1.1 Searching for fixed points

The only fixed point for a given F was found to be x_F as seen in the plot for n=4. The color of the scatter plots is given by $min(Re(\lambda)$s. We know that for a Hopf bifurcation, the real parts of eigenvalues go to 0 at the bifurcation. Thus, we can graphically see the bifurcation point for $n = 4$ (the blue point corresponding to $min(Re(\lambda)) = 0$). Moreover, the only fixed point seen is x_F (evidenced by $x_{1_{eq}}$ v/s F being on the line $y = x$)



(a) Approximate location of fixed points. (Color map is the value of $||f(x)||$)



(b) Hopf bifurcation seen for n=4 for $F > 0$.

NOTE: The following codes are found in the `bifurcations` directory.
(1) The first code to be run is `finding_fixed_pts.py`, that finds the fixed points for a given (n, F) and saves the dataset. The location plots on the LHS is obtained by uncommenting a portion of this code.
(2) For all $(F > 0)$ cases, it is repeatedly seen that the only fixed point obtained is $x_F = (F, F, \ldots \text{n times})$. This is seen in the plot on the RHS, where $x_1FP = F$. The Hopf bifurcation is identified. This plot is obtained by running `hopf_bifurcation_plots.py`

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3.1.2 Eigen value analysis of x_F

For different dimensions, eigenvalues of x_F calculated for a range of F. Points at which $Re(\lambda)) = 0$ would be Hopf bifurcations. If $Re(\lambda) = 0$ for more than one pair of lambdas, it would be a Hopf-Hopf bifurcation. Thus, could get the following table of F_H (values of F at the first Hopf bifurcations) as a function of n:

n	Numerical Estimate		Analytic Estimate	From Van Kekem
	Hopf	Hopf-Hopf		-2018
4	0.994974874	-	1	1
5	0.889447236	-	0.894427190999916	0.8944272
6	0.994974874	-	1	1
7	1.175879397	-	1.18201809701221	1
8	0.994974874	-	1	-
9	0.889447236	-	0.898197570222574	-
10	0.889447236	-	0.894427190999916	-
11	0.934673367	-	0.934338653241817	-
12	-	0.994974874	1	-

Figure 2: Table containing F_H values of the first Hopf/ Hopf-Hopf bifurcations as a function of n. Compared with reference paper's numerical and analytic results.

This is obtained by analysing the eigen values of x_F. This table is generated upon running `hopf_table.py`.
Runtime = 2.576 seconds

3.2 $F < 0$

Here, searching for fixed points to elucidate the precise bifurcation structures is the most important aspect. This is done by running the `finding_fixed_pts.py` for differently sized grids/ resolutions for different dimensions, to generate the datasets (this must be run before the remaining codes). The runtimes are also rather large:
a. $(n = 4)$: 23.897 seconds
b. $(n = 5)$: 987.456 seconds
c. $(n = 6)$: 1674.324 seconds

We search for fixed points extensively here. We may have up to 2 pitchfork bifurcations; each bifurcation would be in a different dimension. Using a color coding to differentiate between stable and unstable fixed points on the bifurcation cure (color = of$\lambda < 0$, darker the color, more stable the point), the curves would look like the following:

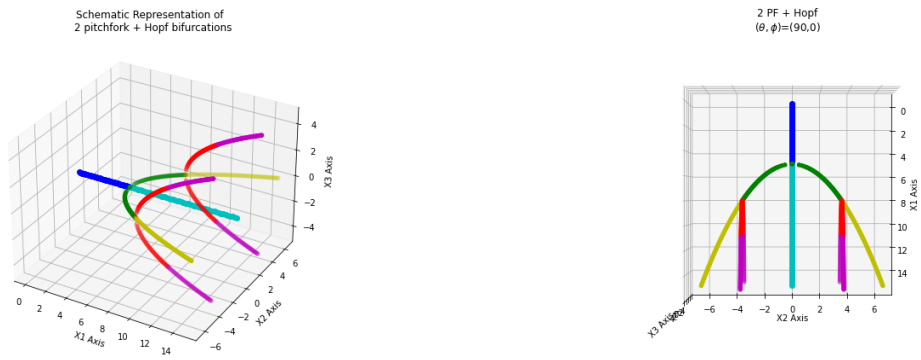


Figure 3: Schematic Diagrams of multiple pitchfork bifurcations

The above diagrams are generated by running `bifurcation_schematic.py`. The subsequent bifurcation diagrams are plotted by running `pf_bifurcation_plots.py`

3.2.1 $n = 4$

x_F undergoes a supercritical pitchfork bifurcation, its two stable branches then go through another supercritical PF. This is followed by the 4 branches undergoing a simutaneous supercritical Hopf bifurcation.

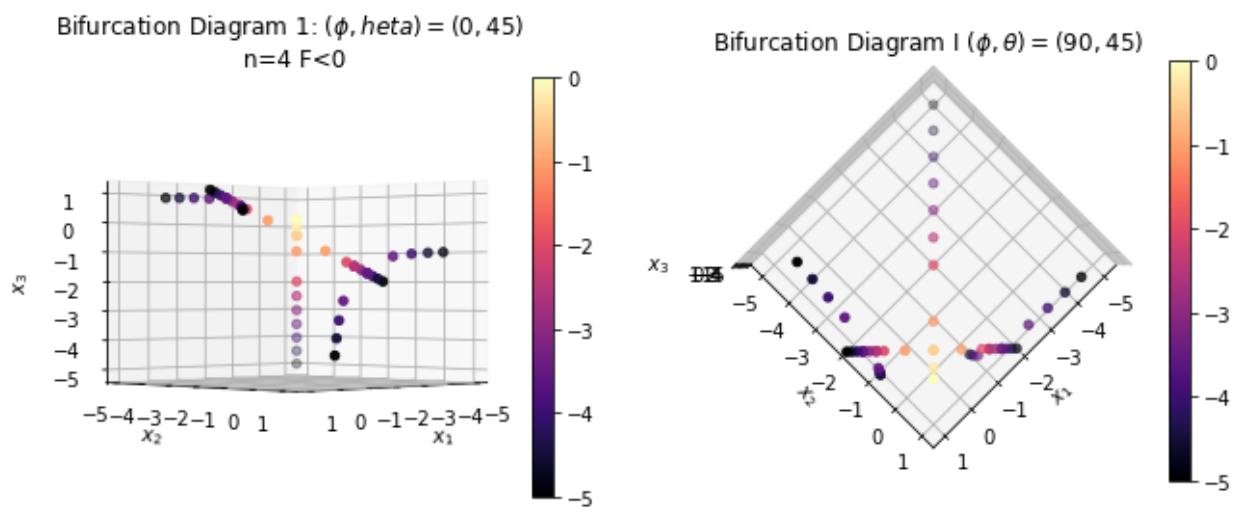


Figure 4: $n = 4$ bifurcation structure. Color corresponds to value of F to make correlation between this and schematic diagrams evident.

Now, to understand the stability, we have another color coding scheme:

$$\begin{aligned} \text{Number of } \lambda\text{s such that}(\lambda > 0) &= 0 \implies \text{no bifurcation} \\ &= 1 \implies \text{pitchfork bifurcation has occured} \\ &= 2 \implies \text{Hopf bifurcation has occured} \end{aligned} \tag{1}$$

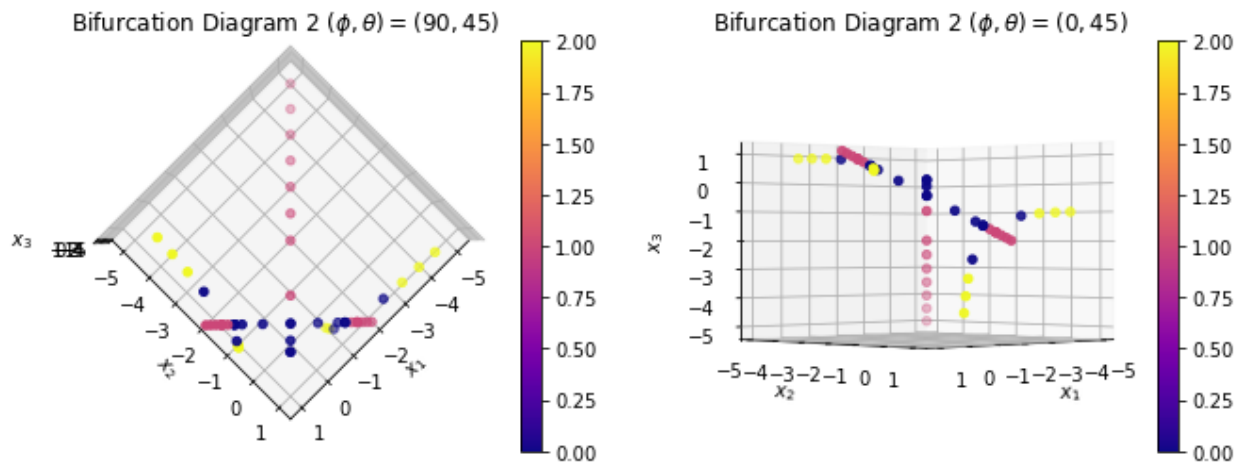
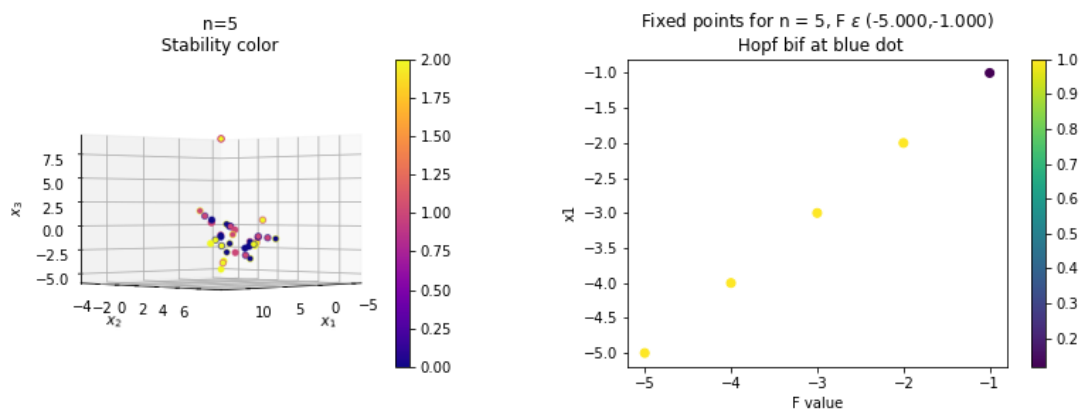


Figure 5: $n = 4$ bifurcation structure. Color corresponds to stability of points, confirming the outlined bifurcation structure

3.2.2 n=5

Though this has multiple fixed points apart from x_F, they are not arising from pitchfork bifurcations (as they are not possible for odd dimensions). Thus, x_F only undergoes Hopf bifurcation which be visualised using the Hopf color scheme (as seen previously).



(a) Pitchfork bifurcaton structure is lost. (b) Hopf bifurcation seen for $F_H = -1$

Figure 6: $n = 5$ bifurcation structure. Color corresponds to stability of points, confirming the outlined bifurcation structure

3.2.3 n=6

As a $4k + 2$ system, x_F undergoes 1 pitchfork bifurcation and the branches undergo Hopf bifurcations.

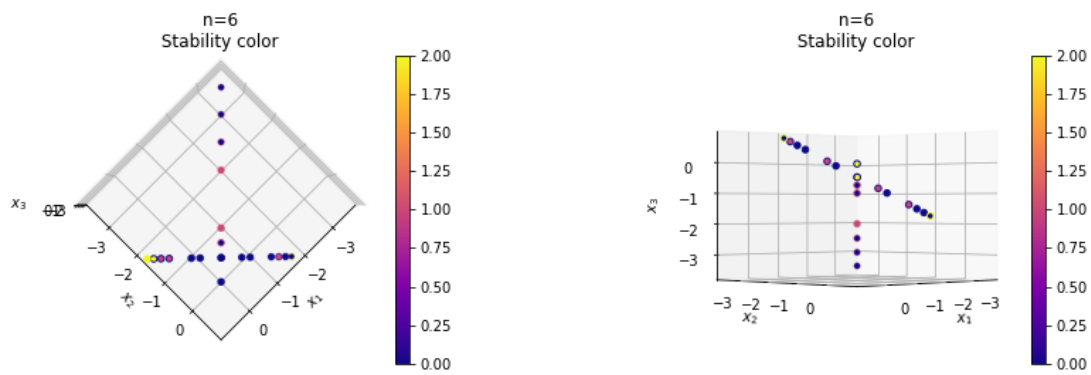


Figure 7: $n = 6$ bifurcation structure. Color corresponds to stability of points.

Thus, the previously outlined bifurcation structures for $n = 4k$ (=4), $n = 2k + 1$ (=5) and $n = 4k + 2$ (=6) are numerically obtained too.

4 Chaotic Behaviour

At first, we would like some estimates on where we can expect chaotic behaviour to kick in. This is done by calculating Lyapunov exponents as a function of F for different dimensions, n. We can get F_{chaos} as the first value of F for which the largest Lyapunov exponent(s) > 0. Thus, we can see how F_{chaos} vary with dimension.

4.1 Calculating Lyapunov Exponents

NOTE: The following codes are found in the `chaos` directory. We must first run `lyaexp_function_of_F.py` to generate the datasets that, for a given dimension n, calculate the Lyapunov spectrum as a function of F over a range of F values. This is again a time consuming code to run, and takes ~ 550 seconds for every dimension, totalling to $7 \times 2 \times 550 \sim 2+$ hours.

ALGORITHM: Calculating Lyapunov Spectrum using QR Methods.

1. We must solve not only the n dimensional system of ODEs but also the n^2 dimensional system of their linearisations (matrix M) simultaneously.

2. After integrating for a period of time, we decompose M into it's QR components (as with time, all the vectors will start aligning alongside the largest Lyapunov vector. Thus, we need to decompose them into orthonormal components). We do this repeatedly for a total time of Δt

3. The Lyapunov exponents are logs of diagonal elements of the matrix R (averaged over all the times we have done the QR factorisation:

$$(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{\sum \log(Diag(R))}{\Delta t}$$

Therefore, we have the following:

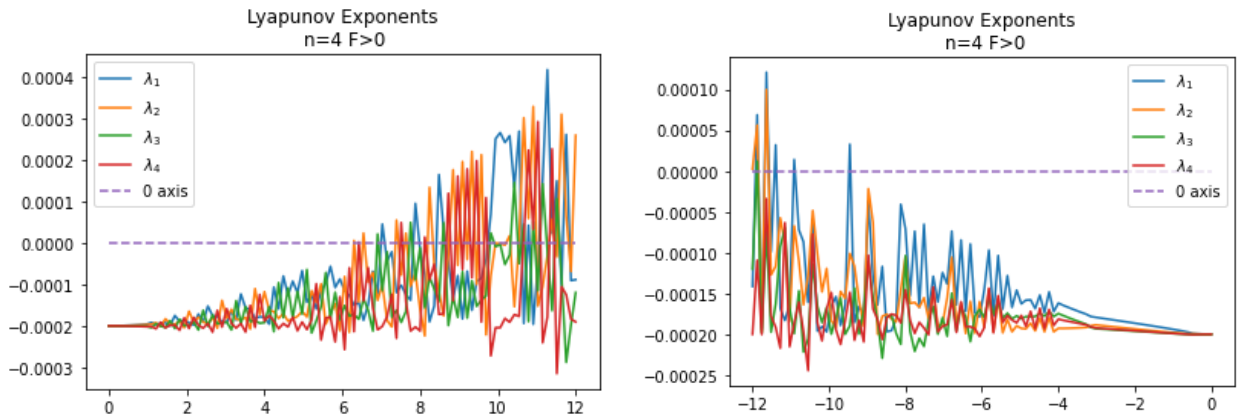


Figure 8: 4 largest Lyapunov exponents as a function of F for $n = 6$

These plots are obtained by running `plotting_lya_exp.py`

We can compile the above information for different dimensions n to give:

F < 0		F > 0	
n	F_chaos	n	F_chaos
4	-0.24242424	4	11.75757576
5	-7.87878788	5	6.363636364
6	-10.4242424	6	7
7	-7.39393939	7	5.656565657
8	-8.72727273	8	6.080808081
9	-6.42424242	9	5.868686869
10	-6.3030303	10	6.01010101

Figure 9: F_{chaos} values as a function of dimension n

The above tables are generated by running `F_critical_table.py`

Van Kekem (2019) plotted F_{chaos} as a function of dimension too. However, their values are quite different from the ones obtained above:

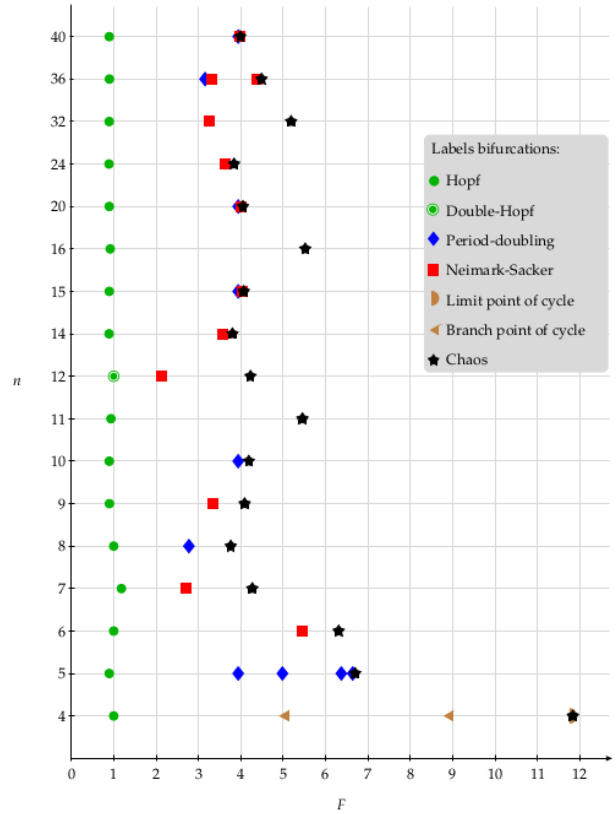


Figure 10: Taken from (van Kekem (2018b)) showing values of F for different bifurcations.

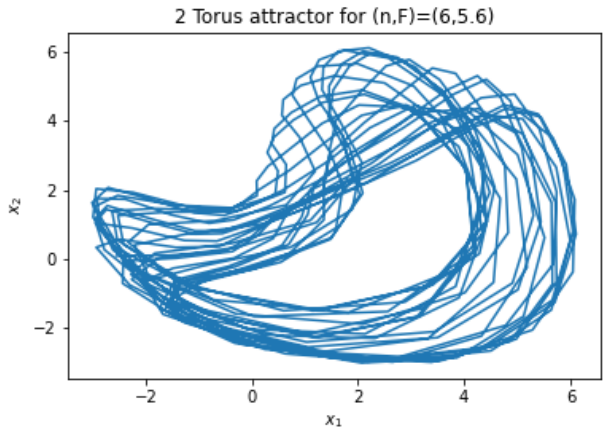
This could be because:
Lyapunov exponents are not robust, reproducible characterustic of a system. Using different algorithms for the exact system can give rise to differing values. Even for systems as widely studied as the Lorenz equations, differing values of Lyapunov exponents have been reported in literature.

4.2 Some routes to Chaos

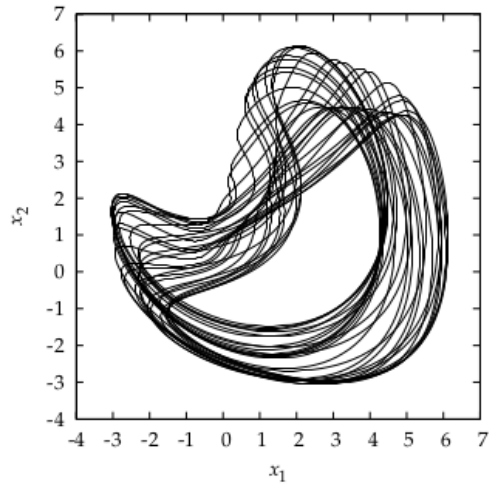
For values of F beyond the regime of the first few bifurcations, we see many interesting behaviours. These are all a part of the cascade of bifurcations that take place, leading to multiple unstable manifolds that ultimately gives rise to chaos.

A helpful tool for finding patterns in the dynamics especially for dynamical systems with high dimensionality are *Poincare sections*. The Fourier transform of the time series of trajectories could also yield helpful information. These two are applied in the following cases:

4.2.1 2-torus attractor (n=6)

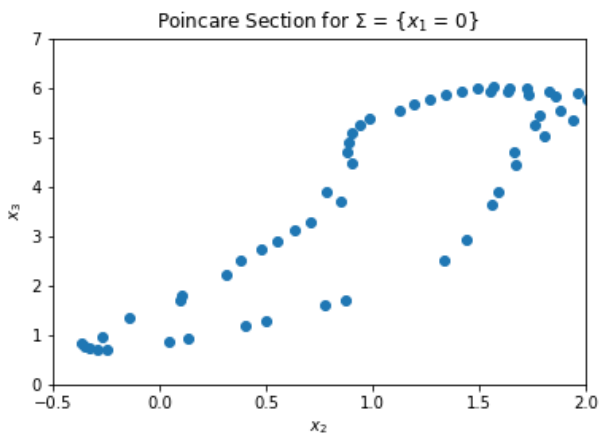


(a) Trajectory plotted

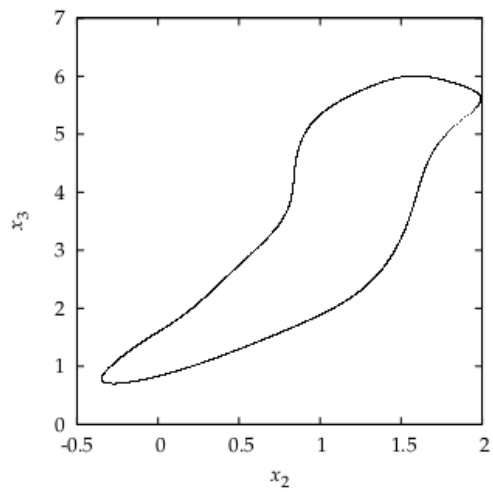


(b) Taken from van Kekem (2018b)

Figure 11: Trajectories for $n = 7$ stay on a 2-torus attractor



(a) Poincare Section



(b) Taken from van Kekem (2018b)

Figure 12: Poincare sections look similar and show an invariant circle.

These plots are obtained by running `chaos_n=6_poincare_sections.py`

4.2.2 Period Doubling bifurcations (n=5)

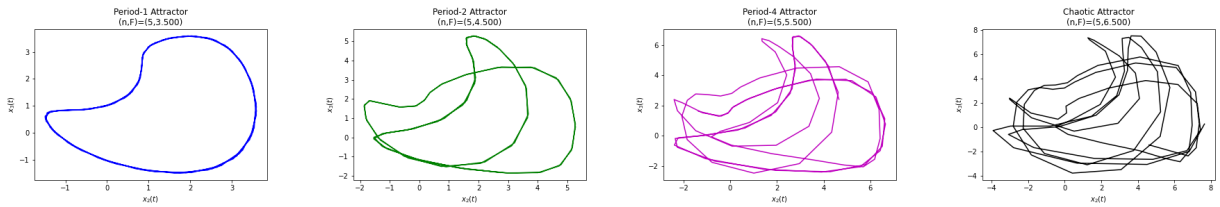


Figure 13: Period doubling bifurcations seen in $n = 5$ systems.

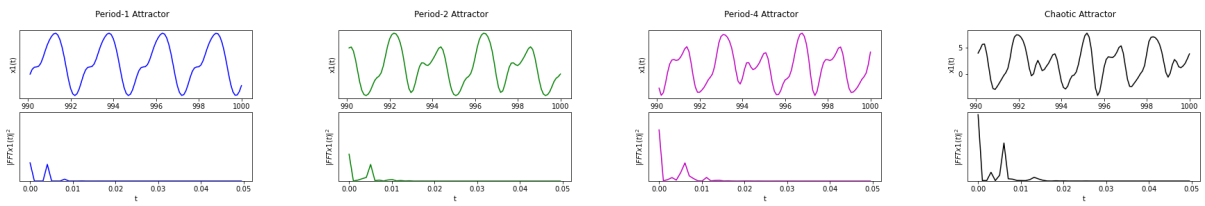


Figure 14: Time series and FFT power spectra of PD bifurcations seen in $n = 5$ systems.

These plots are obtained by running `chaos_n=5_period_doubling.py`

5 Conclusions and Future Directions

Thus, we have scratched the surface of understanding the dynamics of the L96 model. We could verify the bifurcation structures, come up with tentative measures for when chaotic behaviour arises in the system and explored some routes towards such chaos. A major future direction is more work on the routes to chaos in $F < 0$ systems which has been largely unstudied so far.

6 References

[1] Pelzer AFG, Sterk AE. Finite Cascades of Pitchfork Bifurcations and Multistability in Generalized Lorenz-96 Models. Mathematical and Computational Applications. 2020; 25(4):78. https://doi.org/10.3390/mca25040078

[2] Lorenz, E.N. (1969a), Atmospheric Predictability as Revealed by Naturally Occurring Analogues, Journal of the Atmospheric Sciences 26 (4), pp. 636646.

[3] van Kekem, D. L. (2018). Dynamics of the Lorenz-96 model: Bifurcations, symmetries and waves. Rijksuniversiteit Groningen.